



Urgency-Aware QoE-Driven Multi-UAV Fleet Trajectory Optimization for Multi-Priority Post-Disaster Networks via MADRL

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Outline

1. Introduction
2. Proposed Solution
3. Performance Results
4. Conclusion

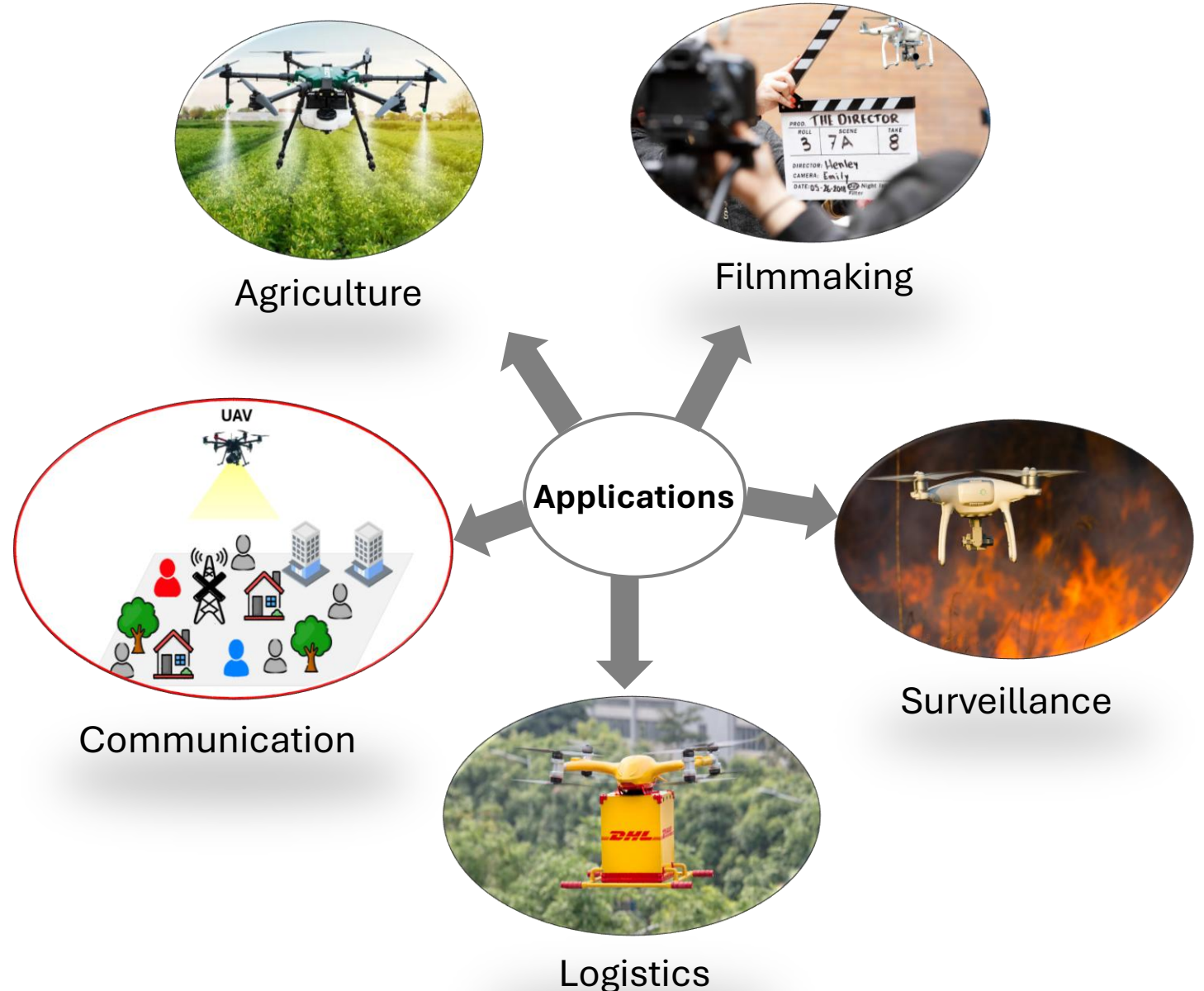
UAVs: Transforming Industries, Expanding Human Capabilities

What are UAVs?

Unmanned Aerial Vehicles (UAVs), known as drones, are remotely controlled or autonomous aircraft.

Why UAVs Matter?

UAVs offer unique capabilities to perform tasks with enhanced speed, safety, and efficiency, especially in remote, or hard-to-reach areas.



Motivation: Enhancing UAV-Assisted Communication



The Critical Problem

- Disasters destroy terrestrial communication infrastructure, creating critical connectivity gaps.
- This loss of coverage severely hinders life-saving emergency response and rescue operations.

UAV-Assisted Communication

- UAVs function as temporary aerial base stations to rapidly restore network connectivity.
- Their high mobility and altitude advantage make them ideal for servicing isolated or heavily damaged zones.



Challenges

Urgency Awareness:

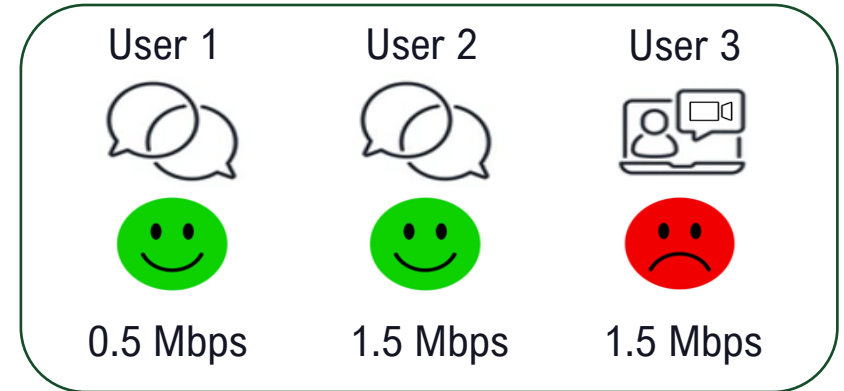
In large-scale disasters, not all users require the same level of priority; dynamic urgency-awareness is essential for saving lives.

Quality of Experience (QoE):

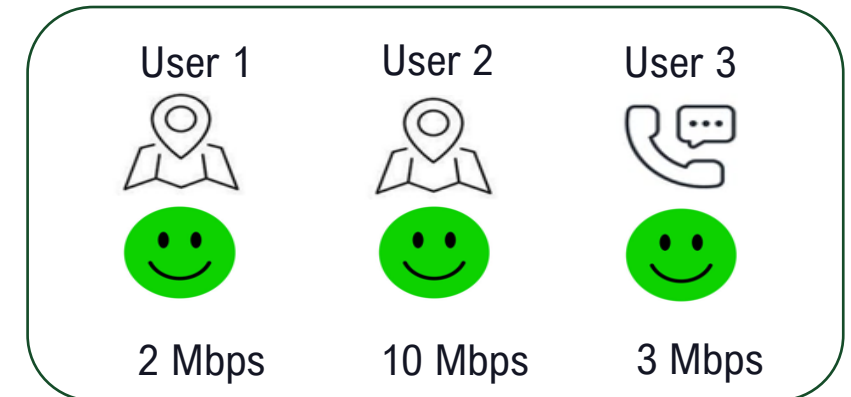
High throughput does not guarantee service satisfaction; QoE modeling is required to capture the subjective utility ground users actually perceive.

Trajectory Design:

Post-disaster urban environments feature dynamic user mobility, unpredictable emergency events, and physical blockages. UAV flight paths directly determine which users are served and when.



(a) A throughput gain is not large enough for some types of service. It makes no difference to a user's QoE.



(b) A larger throughput would be meaningless when a user's QoE cannot be improved anymore.

QoE model using Mean Opinion Score (MOS)

Objective: Maximize user satisfaction via demand-centric resource management [8].

1. Logarithmic Mapping (Continuous score x): The user data rate $R_{n,t}$ is mapped to a continuous quality score $x_{n,t}$

$$x_{n,t} = a \cdot \log_{10} \left(\frac{R_{n,t}}{b} \right)$$

where $a = \frac{x_2 - x_1}{\log_{10}(\theta_2/\theta_1)}$ $b = \frac{\theta_1}{10^{x_1/a}}$ • Range: $\theta_1 = 0.5$ Mbps (MOS 1) to $\theta_2 = 5.0$ Mbps (Saturation, MOS 5)

2. Discrete Mapping to MOS: MOS is defined on a 1–5 scale to reflect human-perceived quality:

$$MOS_{n,t} = \begin{cases} 0, & R_{n,t} \leq R_{th} \\ 1, & R_{n,t} > R_{th} \text{ and } x_{n,t} < 1.5 \\ 2, & 1.5 \leq x_{n,t} < 2.5 \\ 3, & 2.5 \leq x_{n,t} < 3.5 \\ 4, & 3.5 \leq x_{n,t} < 4.5 \\ 5, & x_{n,t} \geq 4.5 \end{cases}$$

- Outage (MOS = 0): $R_{n,t} \leq 100$ kbps (Hardware limit).
- Discrete Scale: 1 (Poor) to 5 (Excellent) based on $x_{n,t}$ thresholds.

Embedding Urgency into Perceived MOS

The Limitation of Standard MOS

- Standard MOS benchmarks network QoE uniformly, failing to differentiate urgency across heterogeneous user classes.
- It ignores the **connectivity relief effect**, where low throughput provides critical satisfaction to emergency nodes but minimal value to standard users.

Introduced Priority Factor ($\alpha_{n,t}$)

- A configurable priority factor to weight user class urgency

$\alpha_{n,t} \in \{\alpha_{\text{Normal}}, \alpha_{\text{Important}}, \alpha_{\text{Emergency}}\}$, In this work, $\alpha_{n,t} = \{1.0, 1.5, 2.0\}$ respectively.

- The priority-aware satisfaction $S_{n,t}$ embeds this urgency directly into the standardized MOS metric:

$$S_{n,t} = \min(\alpha_{n,t} \text{MOS}_{n,t}, 5)$$

Note: Priority weights (1.0, 1.5, 2.0) are open input parameters. Empirical survey data can be directly plugged in to reshape the satisfaction curves without altering the underlying optimization framework.

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Research Gap

- Existing literature heavily focuses on network-centric and uniform QoE optimization, failing to account for user heterogeneity and urgency in post-disaster communication networks.
- Some current priority-aware methods rely on rigid classifications, lacking dynamic spatial adaptability (e.g., binary IoT classifications).
- In summary, priority-aware, QoE-driven trajectory optimization for multi-UAV fleets in 3D urban environments has not been fully addressed in a single framework.



This thesis bridges these gaps, combining a 3D urban environment with a priority-aware, QoE-driven MADRL trajectory framework.

Reference	3D Env.	Multi-UAV	RL/MARL	QoE/MOS	Priority-Aware
Chowdhury et al. [2]					
Cherif et al. [3]	✓				
Hoang et al. [4]			✓		
Hoang et al. [5]		✓	✓		
Ruan et al. [6]		✓			
Guan et al. [14]		✓	✓		
Hoang et al. [7]	✓	✓	✓	✓	
Dai et al. [8]		✓		✓	
Wan et al. [11]	✓	✓	✓		
Xue et al. [10]	✓		✓		
Pan et al. [9]	✓	✓	✓		
Hoseini et al. [18]			✓		✓
Hu et al. [19]					✓
Our Work (Phase 1) [12]			✓	✓	
Our Work (Phase 2)			✓	✓	✓
This Thesis	✓	✓	✓	✓	✓

Proposed Solution

1

Deploy UAVs as flexible aerial base stations

2

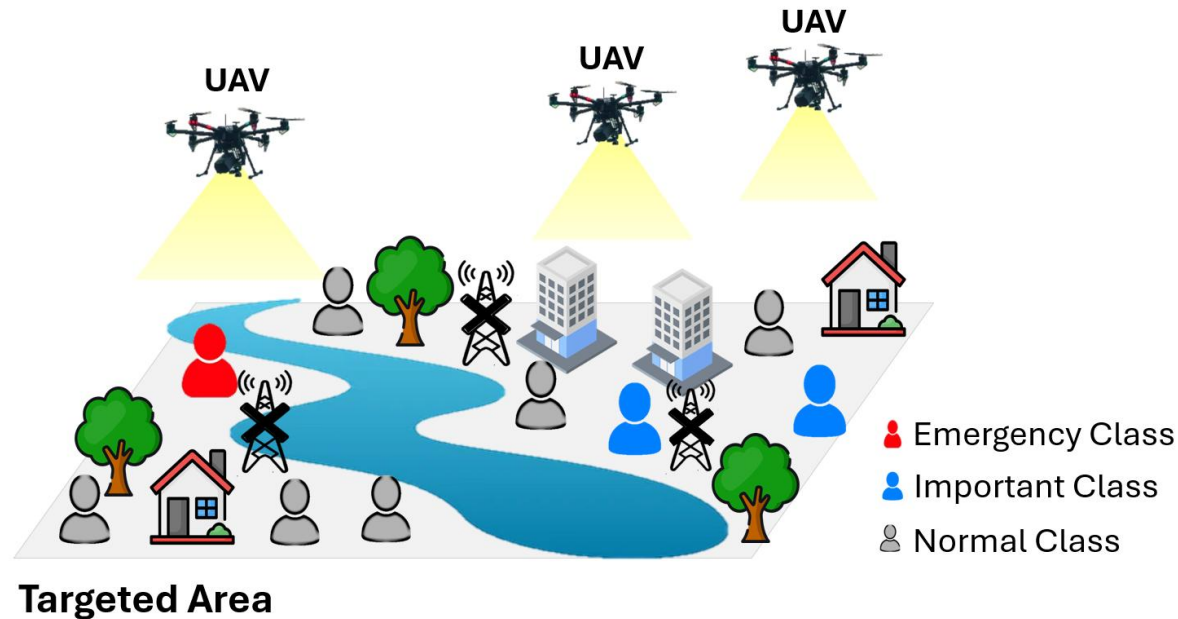
Maximize user Quality of Experience instead of raw throughput

3

Utilize an urgency-aware utility framework to dynamically prioritize critical users

4

Train the fleet using Multi-Agent Deep Reinforcement Learning (MADRL) to learn optimal 3D trajectories in real-time.



Problem Formulation

1. **Objective Function:** To optimize the 3D trajectories of the UAV fleet to maximize the collective priority-weighted satisfaction of ground users over the mission duration.

$$\mathcal{P}: \max_{\{\mathbf{q}_{m,t}\}} \sum_{t=1}^T \sum_{n=1}^N w_{n,t} S_{n,t}$$

$S_{n,t}$ is the priority-scaled satisfaction score: $S_{n,t} = \min(\alpha_{n,t} \text{MOS}_{n,t}, 5)$

$w_{n,t}$ is the class-dependent optimization weight assigned to user n at time t , to establish an optimization priority hierarchy, $w_{\text{Emergency}} > w_{\text{Important}} \gg w_{\text{Normal}}$

2. 3D Kinematics, Beamforming, Service Time Constraints:

- Mobility: $\|\mathbf{q}_{m,t+1} - \mathbf{q}_{m,t}\| \leq v_{\max} \Delta t, \forall m, t$
- Boundary: $\mathbf{q}_{m,t} \in [X_{\min}, X_{\max}] \times [Y_{\min}, Y_{\max}] \times [Z_{\min}, Z_{\max}], \forall m, t$
- Coupled Beamwidth: $\theta_{m,t} = f(z_{m,t}), \Rightarrow z \in [100m, 250m] \rightarrow \theta \in [60^\circ, 120^\circ]$
- Service Continuity: High-priority users (Emergency/Important) are protected by a minimum duration threshold ($T_{\min}^{c_n}$) to prevent user service starvation.

Hybrid Deployment Strategy

Two-Stage Operational Design: Integrates classical heuristic clustering for initialization with deep reinforcement learning for active trajectory optimization.

- **Macro-Deployment (K -Means):** Computes ground user coordinates at episode reset to calculate optimal initial entry points for immediate network coverage.
- **MADRL Service (MAPPO):** Takes full autonomous control over the flight horizon, navigating dynamically to maximize priority-weighted user satisfaction.

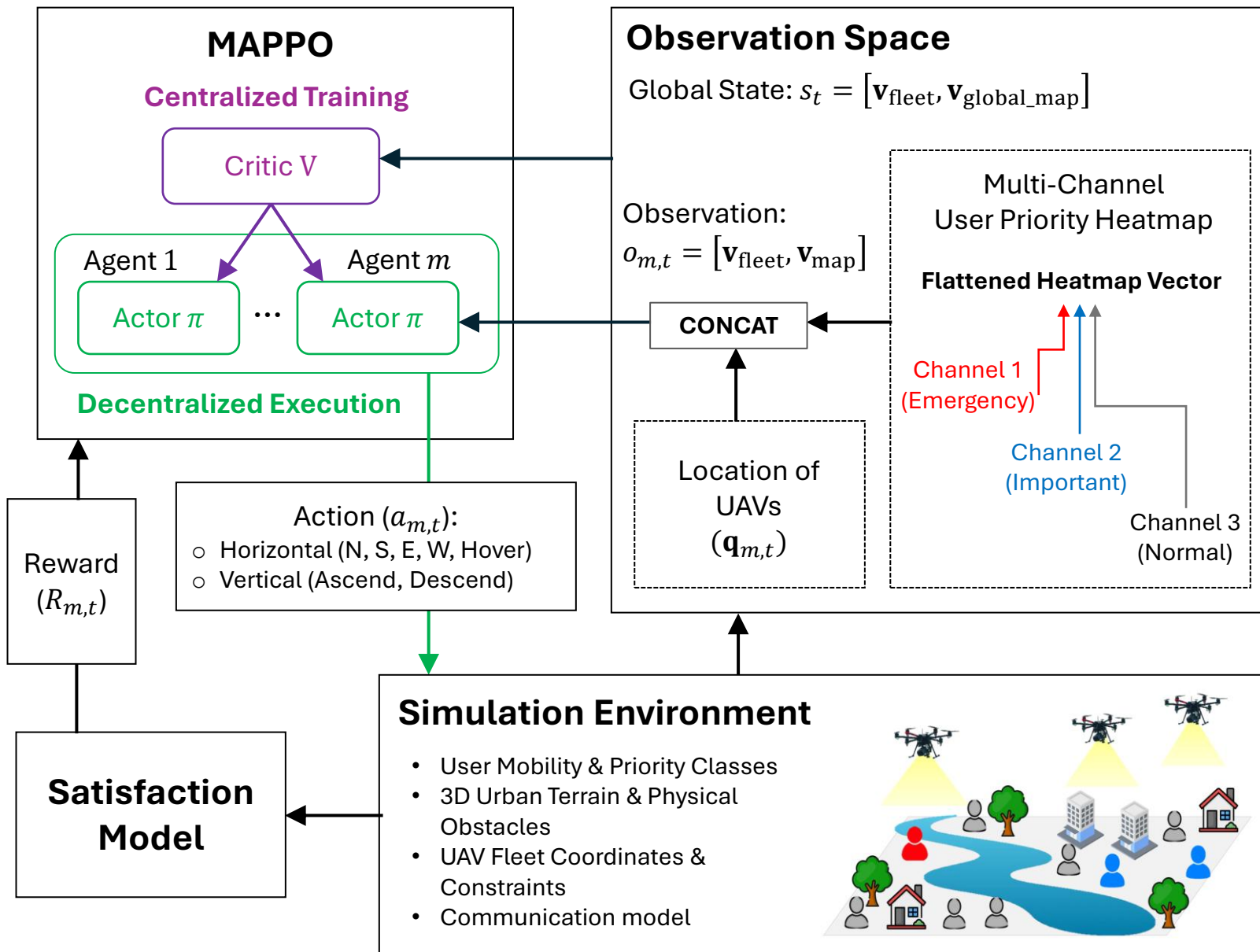
Purpose: Accelerates fleet deployment and stabilizes multi-agent policy convergence by bypassing early training inefficiencies and avoiding empty coverage states.

Algorithm 4.1 Hybrid K -Means Macro-Deployment and MAPPO Training Framework

Require: Maximum episodes E , episode horizon T , update epochs K_{epoch} , minibatch size B , UAV fleet size M

```
1: Initialize shared actor policy  $\pi_\theta$  and centralized critic  $V_\phi$ 
2: for episode  $e = 1$  to  $E$  do
    ▷ Stage I:  $K$ -Means-Based Macro-Deployment (Environment Reset)
3:   Apply  $K$ -Means clustering [15] to ground UE locations
4:   Assign each UAV  $m \in \mathcal{M}$  to a cluster centroid target
5:   Deploy UAVs to initial coordinates and observe initial local states  $\{o_{m,0}\}_{m=1}^M$ 
    ▷ Stage II: Decentralized MAPPO Interaction (Active Control)
6:   Initialize replay buffer  $\mathcal{D} \leftarrow \emptyset$ 
7:   for  $t = 0$  to  $T - 1$  do
8:     for each UAV agent  $m = 1$  to  $M$  do
9:       Observe local observation  $o_{m,t}$ 
10:      Select action  $a_{m,t} \sim \pi_\theta(\cdot | o_{m,t})$ 
11:    end for
12:    Execute joint action  $\mathbf{a}_t = [a_{1,t}, \dots, a_{M,t}]^T$ 
13:    Observe localized rewards  $\{R_{m,t}\}_{m=1}^M$  and next observations  $\{o_{m,t+1}\}_{m=1}^M$ 
14:    for each UAV agent  $m = 1$  to  $M$  do
15:      Store transition  $(o_{m,t}, a_{m,t}, R_{m,t}, o_{m,t+1})$  in shared buffer  $\mathcal{D}$ 
16:    end for
17:  end for
    ▷ Stage III: Centralized Policy Update
18:  Compute returns  $\hat{R}_{m,t}$  and advantages  $\hat{A}_{m,t}$  using GAE and centralized critic  $V_\phi$ 
19:  for epoch  $k = 1$  to  $K_{\text{epoch}}$  do
20:    Sample minibatches of size  $B$  from shared buffer  $\mathcal{D}$ 
21:    Update actor policy parameters  $\theta$  using PPO clipped objective [13]
22:    Update centralized critic parameters  $\phi$  using value loss function
23:    Apply entropy regularization  $\mathcal{H}(\pi_\theta)$  to encourage exploration
24:  end for
25: end for
```

MADRL Training Loop



MADRL: Multi-Agent Deep Reinforcement Learning
MAPPO: Multi-Agent Proximal Policy Optimization
CTDE: Centralized Training with Decentralized Execution

Reward Function Design

Composite Training Reward:

$$R_{m,t} = R_{\text{QoE},m,t} + R_{\text{guide},m,t} - P_{\text{delay},m,t} - P_{\text{overlap},m,t} - P_{\text{collision},m,t}$$

Weight Tuning Optimization:

$$\lambda_{\text{collision}} \gg \lambda_{\text{QoE}} > \lambda_{\text{overlap}} > \lambda_{\text{delay}} > \lambda_{\text{guide}}$$

Establishes physical fleet safety as the dominant soft-constraint, while spatial guidance functions as an auxiliary gradient.

QoE Credit (R_{QoE})

- 65% Global Term: Drives joint fleet-wide cooperation.
- 35% Local Term: Maintains accountability; avoids hotspot grouping.

Guidance Soft-Claim (R_{guide})

- Pulls agents toward high-deficit clusters.
- 400m Coordination Zone: Suppresses redundant attraction to prevent dogpiling.

Emergency Service Delay (P_{delay})

- Accumulates consecutive unserved steps for Emergency users.
- Tightly restricted to nearest UAV for active Emergency users.

Safety Spacing ($P_{\text{overlap}}, P_{\text{collision}}$)

- Overlap Margin ($\delta = 0.8$): Grants a flexible 20% footprint handover buffer.
- $P_{\text{collision}}$: Enforces a mandatory 50m physical separation distance.

Curriculum Learning Strategy

A three-phase curriculum that progressively scales complexity to stabilize training.

Phase 1: Basic Navigation (Cooperative Flight)

Environment: Fixed altitude ($z_{\text{fixed}} = 120m$), free-space path loss (no blockages).

Objective: Learn basic horizontal tracking and cooperative anti-clustering.



Phase 2: 3D Pathfinding & Beamforming (Vertical Actions)

Environment: Variable altitude $[z_{\text{min}}, z_{\text{max}}]$ with vertical actions.

Objective: Learn the altitude-coverage trade-off:

- High altitude: Wide footprint for scattered users.
- Low altitude: Narrow beam to maximize SNR and QoE.



Phase 3: Complete Urban Environment (Signal Blockage Mitigation)

Environment: 3D urban terrain with building blockages.

Objective: Learn to maintain Line-of-Sight (LoS) links and mitigate building shadows.

Simulation Settings

Mission: Deploy 3 UAVs to cooperatively maximize priority-weighted QoE for 150 mobile ground users in a $2 \times 2 \text{ km}$ 3D urban disaster area, balancing fleet-wide coverage with rapid response to critical nodes.

ENVIRONMENT & UAV

Simulation area

2000 × 2000 m, bound $[-1000, 1000]$

UAV configuration

$M = 3$, $v_{\max} = 25 \text{ m/s}$

Antenna beamforming

60° – 120° coupled to altitude 100–250m

CHANNEL MODEL

Carrier & bandwidth

5.8 GHz, using full 20 MHz,
Synchronized ICIC

Transmit power

25 dBm (UAV downlink)

Noise floor

−90 dBm thermal noise

Propagation model

3D FSPL + NLoS shadowing, PL exp
3.2, Rician $k = 50$

USER TRAFFIC & TRAINING

User mobility

150 users, $v = 1.5 \text{ m/s}$ Gaussian random
walk, Hotspot spread $\sigma = 450 \text{ m}$

DRL training

K-means + MAPPO-based CTDE, total
curriculum trains of 20,000 episodes

Disaster wave schedule

Initial Normal users only →
Important → Emergency → Mixed all

PRIORITY MIX & REWARD WEIGHTING

Normal — 94% of users

Priority factor α : 1.0
QoE optimization weight w : 0.1

Important — 4% of users

Priority factor α : 1.5
QoE optimization weight w : 1.5

Emergency — 2% of users

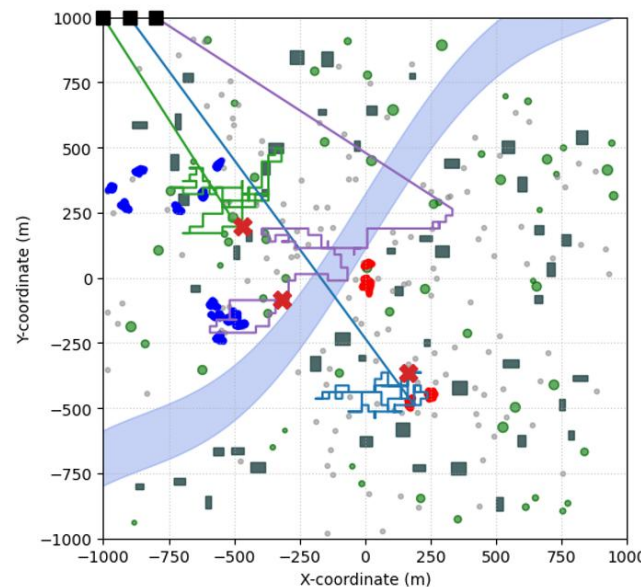
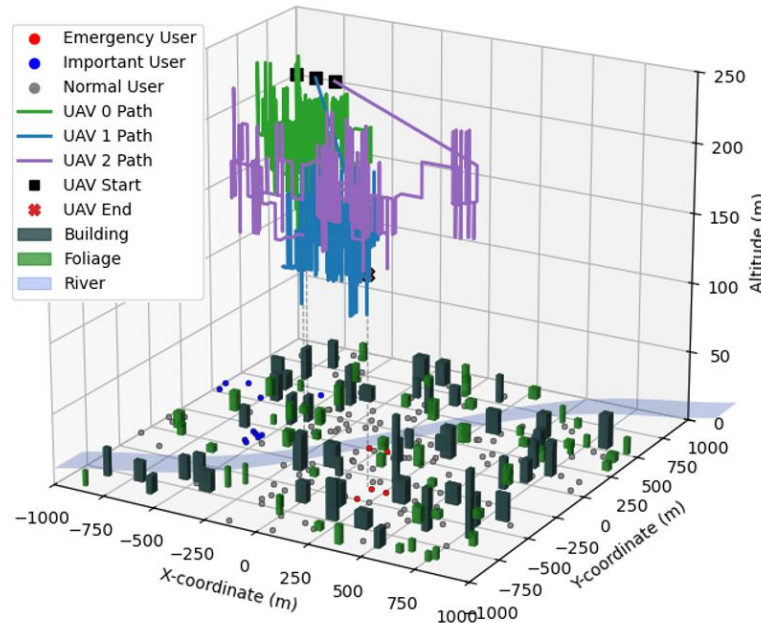
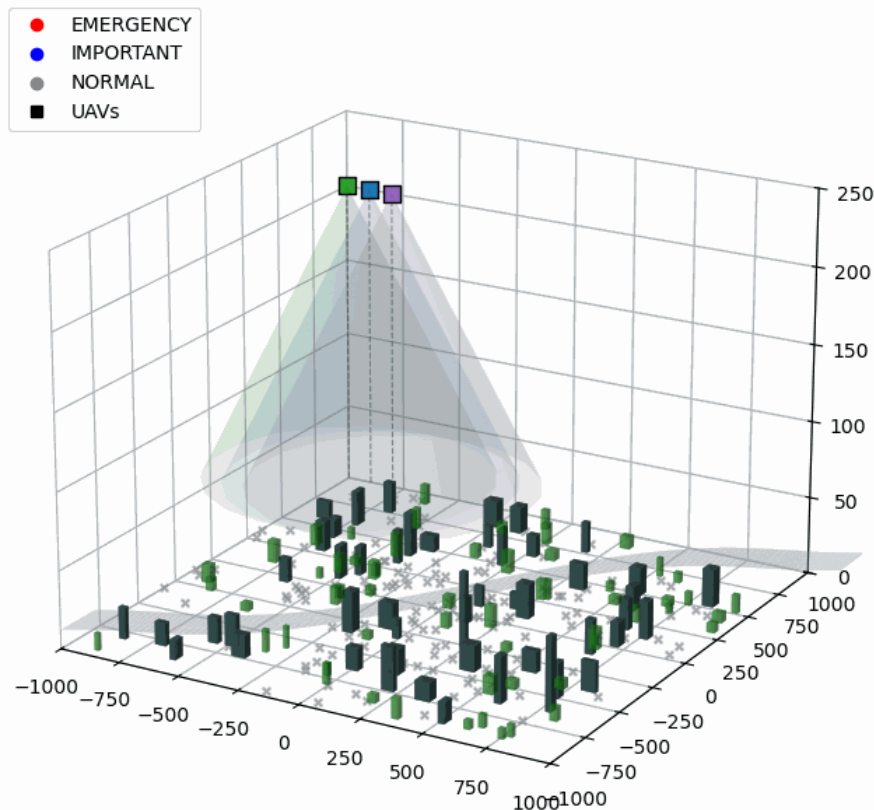
Priority factor α : 2.0
QoE optimization weight w : 3.0

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Results (1): Trajectory and Behavior

3D Trajectory & Adaptive Beamforming Cones (Step 0)



Two-Stage Deployment:

- Straight-line flight from corners to K-Means centers, followed by active MAPPO control.

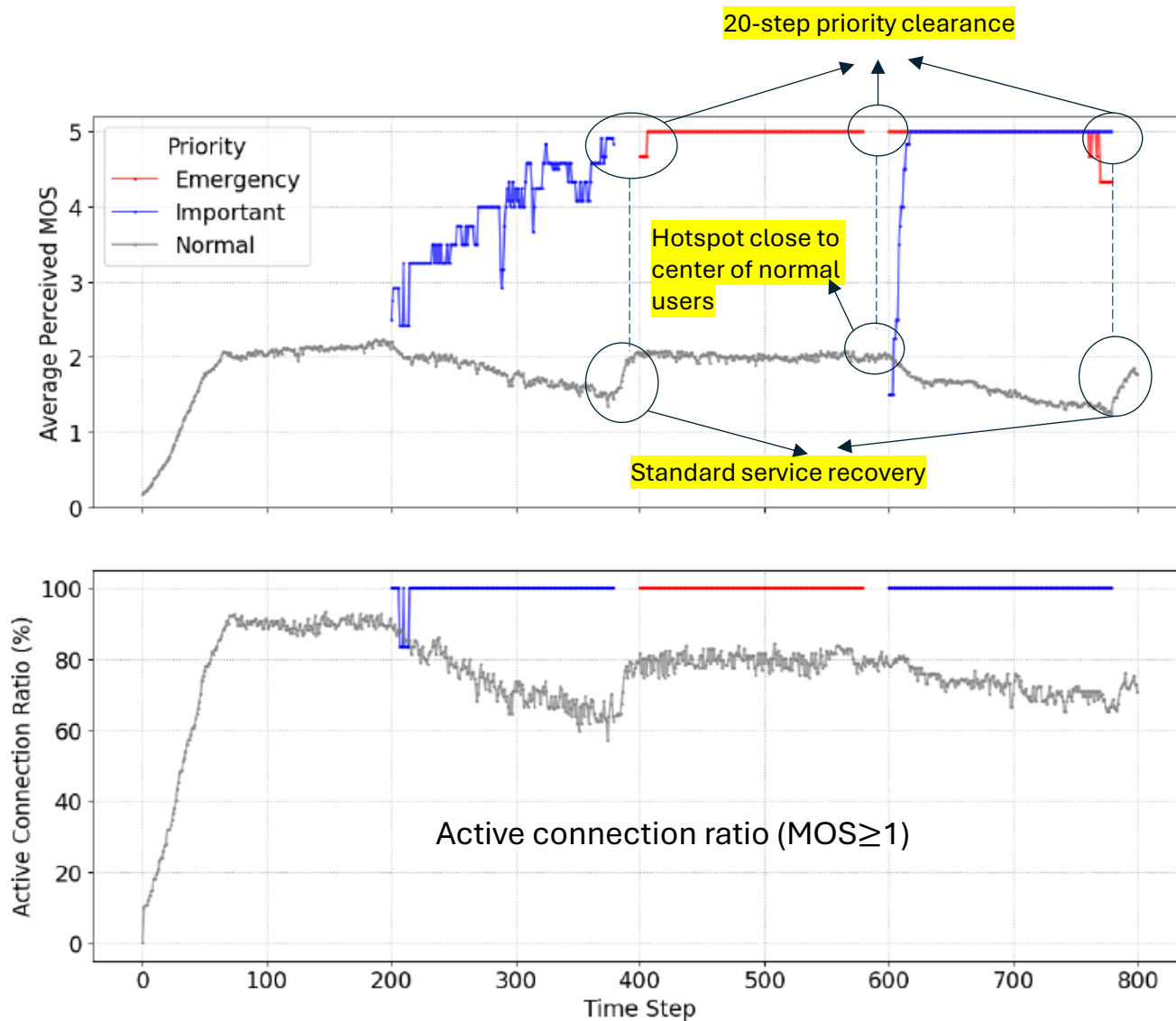
3D Altitude & Beamwidth Adaptation:

- **Ascend:** Expands footprint for user search and mitigate building blockages.
- **Descend:** Focuses antenna beams to maximize SNR and user QoE over dense clusters.

Cooperative Spatial Division:

- UAVs partition the environment to maximize coverage while minimizing overlap.

Results (2): Temporal Satisfaction



- **Dynamic Temporal Adaptation ($t = 200, 400, 600$):** UAVs dynamically adjust trajectories and altitudes to prioritize critical clusters upon activation, delivering high-satisfaction outcomes.
- **Standard Service Recovery (Clearance Windows):** UAVs naturally return to tracking the center of mass of Normal users when critical waves clear.

Results (3): Statistical Benchmarking

Objective: Maximize cumulative priority-weighted QoE: $\sum_t \sum_n w_{n,t} S_{n,t}$

Table: 10-Seed Performance Comparison Under Dynamic 3D Urban Environment

Method	Weighted QoE Utility (Points)	Delivered MOS (Quality)			Wait Time (Latency, steps)		Coverage (MOS ≥ 1, %)		
		Emergency	Important	Normal	Emergency	Important	Emergency	Important	Normal
Static Centroid (Baseline)	33037.2 ± 4472.3	1.30 ± 0.87	1.59 ± 0.96	1.49 ± 0.11	122.57 ± 95.33	86.02 ± 52.43	47.2% ± 31.4%	48.9% ± 13.8%	65.1% ± 3.1%
Priority-Agnostic ($\alpha_{n,t} = 1.0, w_{n,t} = 0.1$)	32702.8 ± 3737.2	1.15 ± 0.60	1.48 ± 1.08	1.69 ± 0.10	60.85 ± 63.24	53.17 ± 37.35	49.8% ± 23.2%	63.2% ± 23.2%	70.7% ± 3.2%
Proposed K-MAPPO	44985.9 ± 2449.3	3.32 ± 0.70	3.47 ± 0.99	1.59 ± 0.08	8.28 ± 7.85	14.13 ± 24.19	93.3% ± 6.7%	90.3% ± 15.6%	66.2% ± 3.7%

Optimized Standard
Coverage Trade-off

Key Findings:

- Proposed K-MAPPO achieves **>36% utility improvement** over baselines.
- Reduces Emergency wait time by 93.2% (down to 8.28 steps).
- Boosts Emergency coverage to 93.3% and delivered MOS to 3.32.
- Maintains a stable 66.2% baseline coverage for Normal users.

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Conclusion

Key Achievements:

- **Urgency-Aware QoE Framework:** Shifted focus to human-perceived MOS satisfaction with priority-scaled utility.
- **Two-Stage K-MAPPO Control:** Combined macro K-Means placement with decentralized MAPPO and curriculum training.
- **Proven Performance Gains:** Achieved +36.2% utility and -93.2% Emergency latency (down to 8.28 steps).

Future Work:

- **UAV Energy Constraints:** Model battery and flight power limits to optimize mission longevity.
- **Adaptive Resource Allocation:** Dynamically distribute communication resources to scale up network capacity.

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Q&A